

Temperature Monitor (with LCD Display)

Arduino Uno + Tinkercad Simulation Project

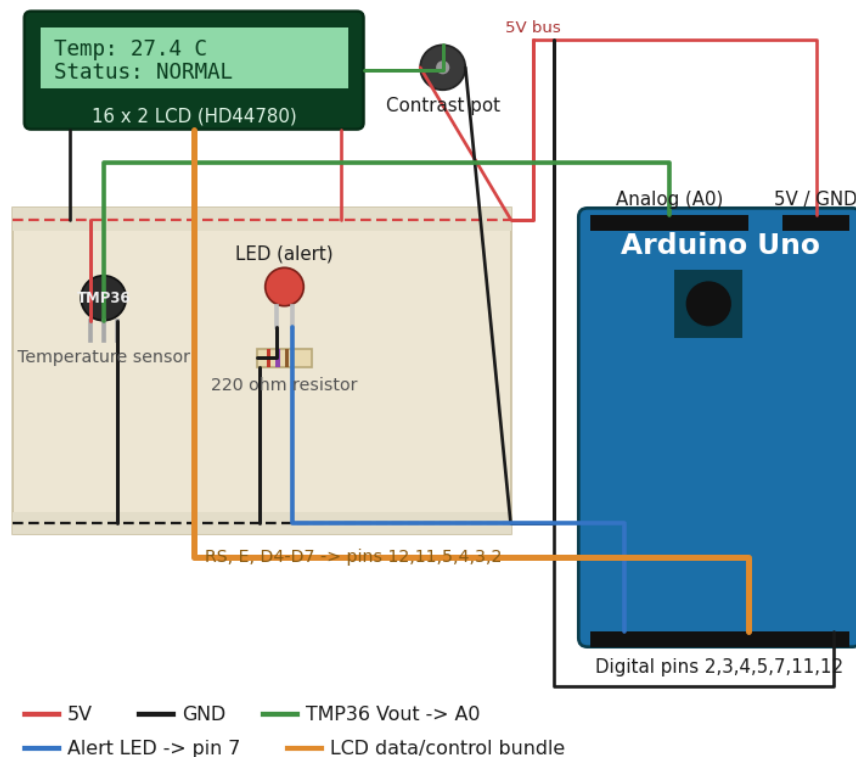
Project Overview

Yeh project TMP36 temperature sensor ka istemal karke ambient temperature ko continuously measure karta hai, aur usse ek 16x2 LCD display par digitally show karta hai. Jab temperature ek fixed threshold (30 C) se upar chala jata hai, ek alert LED automatically ON ho jati hai aur LCD par status "HIGH!" show hota hai. Roshni ya button ki zaroorat nahi - sab kuch Arduino code mein analogRead aur if/else condition se control hota hai.

Components Used

#	Component	Quantity	Purpose
1	Arduino Uno R3	1	Main microcontroller board
2	Breadboard (small)	1	To hold and connect components
3	Temperature Sensor (TMP36)	1	Senses ambient temperature
4	16x2 LCD (HD44780)	1	Displays live temperature and status
5	Potentiometer - 10k ohm	1	Controls LCD contrast
6	LED	1	Alert indicator for high temperature
7	Resistor - 220 ohm	1	Current-limiting resistor for the LED
8	Jumper wires	as needed	Connections between components

Circuit Diagram (Tinkercad Style)



TMP36 wired to A0 (green), alert LED to pin 7 (blue), LCD data/control lines to pins 2,3,4,5,11,12 (orange), and potentiometer wiper to LCD V0. Red = 5V, Black = GND.

Step-by-Step Build Procedure

Step 1: Naya circuit banao

Tinkercad kholo, "Circuits" section mein jao, aur "Create new Circuit" par click karo.

Step 2: Components add karo

Right-side panel se ye cheezein drag karke canvas par lao: Arduino Uno R3, Breadboard (small), Temperature Sensor (TMP36), 16x2 LCD, Potentiometer, LED, aur Resistor (220 ohm).

Step 3: TMP36 sensor wiring karo

TMP36 ka Power pin Arduino ke 5V se, GND pin Arduino ke GND se, aur Vout pin Arduino ke A0 (analog) pin se wire karo.

Step 4: Alert LED aur resistor wiring karo

LED ka anode (lambi leg) Arduino ke pin 7 se wire karo. LED ka cathode (choti leg) 220 ohm resistor ke through GND se jodo.

Step 5: LCD wiring karo

LCD ke VSS aur RW pins GND se, VDD pin 5V se jodo. RS ko pin 12, Enable ko pin 11, aur D4-D7 ko pins 5, 4, 3, 2 se wire karo. Backlight ke LED+ aur LED- pins 5V aur GND se jodo.

Step 6: Contrast potentiometer wiring karo

Potentiometer ke dono side terminals 5V aur GND se jodo, aur middle wiper terminal ko LCD ke V0 pin se wire karo - yeh LCD ka contrast set karta hai.

Step 7: Code likho

Arduino ko select karo, "Code" button dabao, "Blocks + Text" ki jagah Text mode select karo, aur neeche diya gaya pura sketch paste karo (dekhein agla section).

Step 8: Simulation chalao

"Start Simulation" button dabao. TMP36 par click karke uska temperature slider adjust karo - LCD par live temperature update hote dekho. Jab temperature 30 C se upar jaye, alert LED ON ho jayegi aur LCD par "HIGH!" show hoga.

Complete Arduino Sketch

Yeh code neeche pura selectable text hai - directly copy karke Tinkercad ya Arduino IDE mein paste kiya ja sakta hai.

```

// Temperature Monitor with LCD Display
// TMP36 -> A0 | Alert LED -> Pin 7 | LCD -> RS12,E11,D4-D7:5,4,3,2

#include <LiquidCrystal.h>

const int sensorPin = A0;
const int ledPin = 7;
const float thresholdTemp = 30.0; // Celsius

LiquidCrystal lcd(12, 11, 5, 4, 3, 2);

void setup() {
  pinMode(ledPin, OUTPUT);
  lcd.begin(16, 2);
  lcd.print("Temp Monitor...");
  delay(1500);
  lcd.clear();
}

void loop() {
  int rawValue = analogRead(sensorPin);
  float voltage = rawValue * (5.0 / 1023.0);
  float temperatureC = (voltage - 0.5) * 100.0;

  lcd.setCursor(0, 0);
  lcd.print("Temp: ");
  lcd.print(temperatureC);
  lcd.print((char)223); // degree symbol
  lcd.print("C ");

  lcd.setCursor(0, 1);
  if (temperatureC > thresholdTemp) {
    lcd.print("Status: HIGH! ");
    digitalWrite(ledPin, HIGH);
  } else {
    lcd.print("Status: NORMAL ");
    digitalWrite(ledPin, LOW);
  }

  delay(1000);
}

```

Wiring Summary

From	To	Wire Color (suggested)
TMP36 Power	Arduino 5V	Red

From	To	Wire Color (suggested)
TMP36 GND	Arduino GND	Black
TMP36 Vout	Arduino A0	Green
LED anode (long leg)	Arduino Pin 7	Blue
LED cathode (short leg)	220 ohm resistor - leg 1	Direct breadboard row
220 ohm resistor - leg 2	GND rail	Black
LCD VSS / RW	GND rail	Black
LCD VDD	Arduino 5V	Red
LCD RS	Arduino Pin 12	Gray
LCD Enable (E)	Arduino Pin 11	Orange
LCD D4 / D5 / D6 / D7	Arduino Pins 5 / 4 / 3 / 2	Blue / Purple / Orange / Teal
Potentiometer wiper	LCD V0	Green
Breadboard GND rail	Arduino GND	Black

Kaise Kaam Karta Hai

TMP36 sensor ambient temperature ko analog voltage mein convert karta hai. Arduino ka A0 pin ye voltage read karta hai, aur code usse Celsius mein convert karta hai formula $(\text{voltage} - 0.5) \times 100$ se, kyunki TMP36 0.5V par 0 C represent karta hai aur har 10 mV rise 1 C ke barabar hota hai.

LiquidCrystal library Arduino ke digital pins ko LCD ke control aur data pins se link karti hai. Har loop cycle mein LCD ki pehli line current temperature ke saath update hoti hai, aur doosri line "NORMAL" ya "HIGH!" status dikhati hai depending on threshold (30 C). Jab temperature threshold se upar jata hai, Pin 7 HIGH ho jata hai aur alert LED ON ho jati hai.

Note: Agar LCD par text blank ya bohat dark nazar aaye to potentiometer ki contrast value adjust karo. Agar LED behavior ulta lage, to code mein threshold comparison (`temperatureC > thresholdTemp`) ko check karo.

Tinkercad Circuit Screenshot

